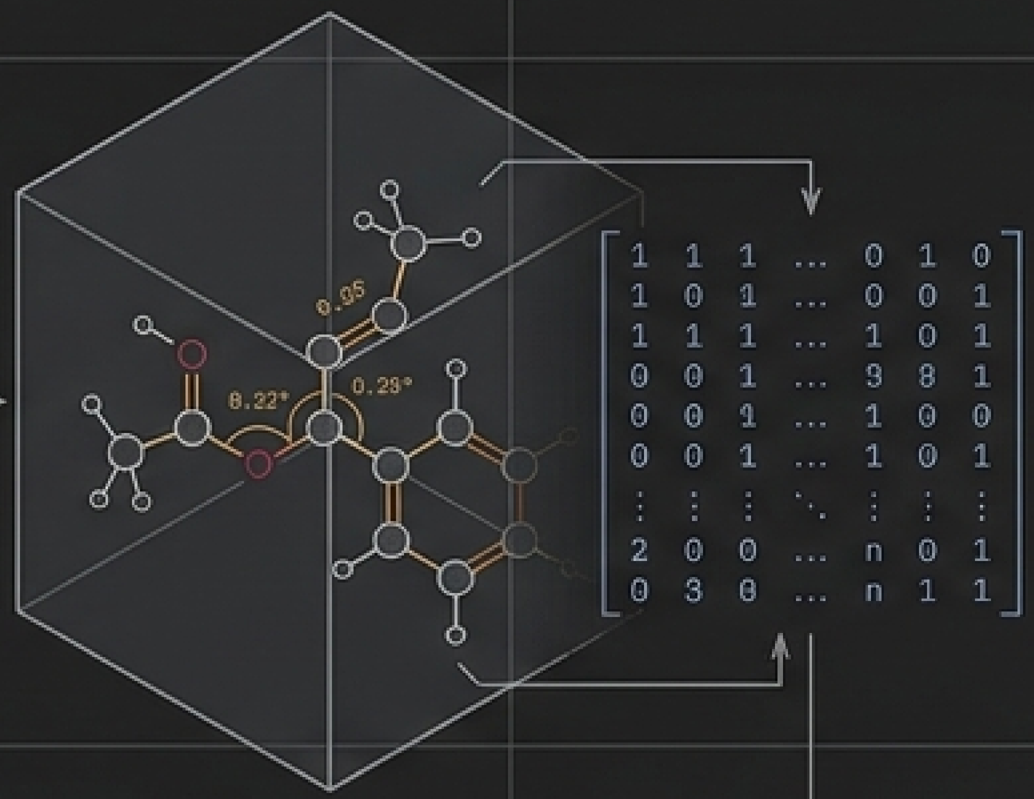
Iso-ACSF: High-Dimensional Invariant Representations for Quantum-Chemical Inference



A feature-engineered manifold transforming QM9 Cartesian geometries into Behler-Parrinello Atom-Centered Symmetry Functions (ACSF).

ORIGIN: QM9 Benchmark Derived | FORMAT: Normalized Array [Samples × Dimensions] | TARGET: Physics-Informed ML Regression

The Quantum-Chemical Modeling Bottleneck

Computational Paradigm Comparison Matrix

Density Functional Theory (DFT)	Raw Cartesian Machine Learning	Iso-ACSF Surrogate Modeling
High absolute accuracy.	Low computational cost.	Low computational cost.
Unscalable computational cost.	Introduces extreme geometric noise.	Mathematically symmetry-invariant.
Severely limits high-throughput screening.	Prone to rotational, translational, and permutational variance.	Projects a physically grounded manifold for stable model inference.

The Feature Construction Pipeline

Raw Input

Cartesian Coordinates (x, y, z) from the QM9 benchmark.

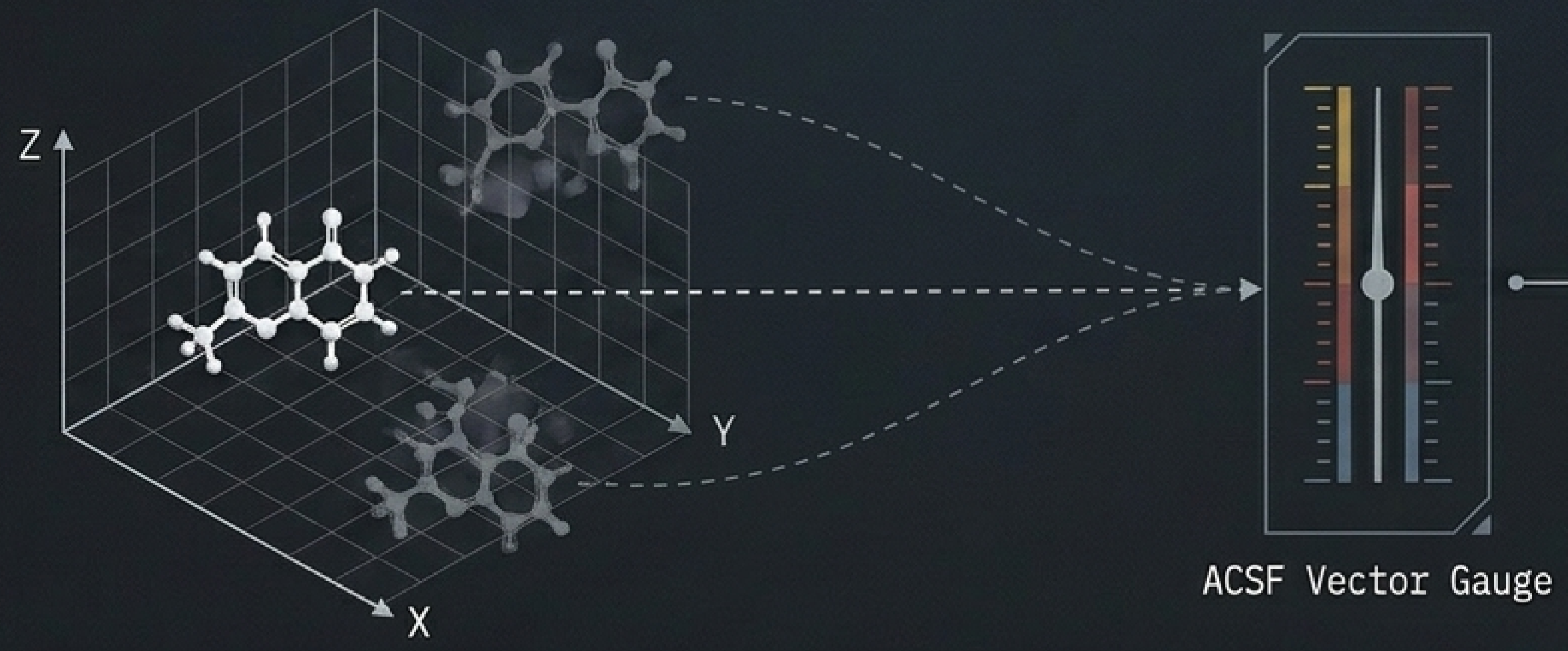
The Transformation Engine

Behler-Parrinello Functions.

The Output Manifold

Iso-ACSF High-Dimensional Descriptor.

Invariance Model



Rotational, translational, and atom-indexing variance is completely eliminated while preserving local atomic environments.

Dataset Anatomy & Tensor Architecture

Column Identity: Each column maps to a uniquely defined, normalized ACSF dimension.

Row Identity: Each row maps to a single, distinct molecular conformation representing a localized chemical state, not just bulk molecular identity.

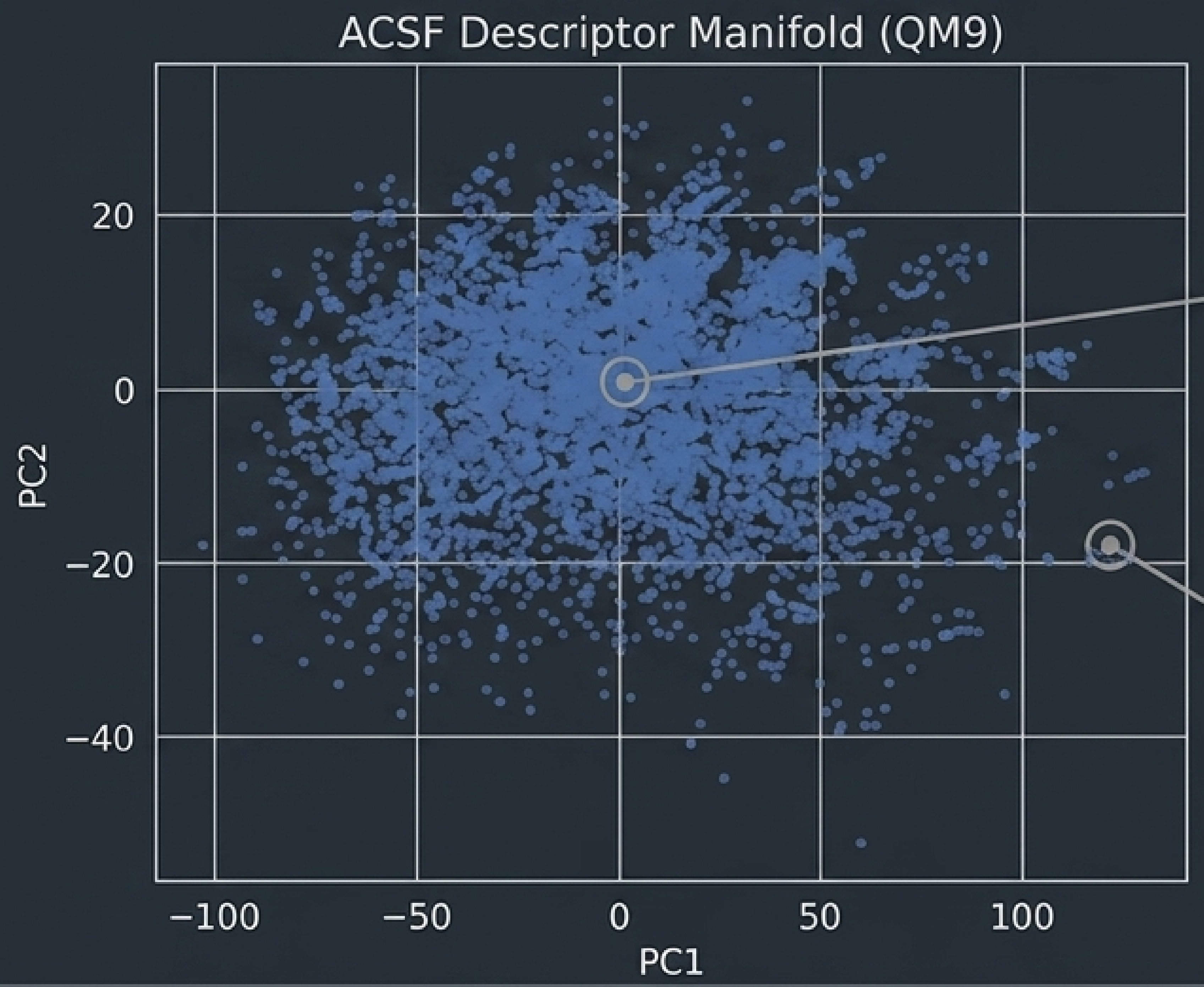
Numerical Stability Parameter

All features are globally normalized across the dataset manifold.

This guarantees structural formatting optimized for gradient-based updates and ensemble-based regression methods.

Reproducibility is strictly guaranteed via deterministic feature generation routines.

State 1: Global Manifold Dispersion



Primary topological density.

Represents baseline stable organic configurations inherited from QM9.

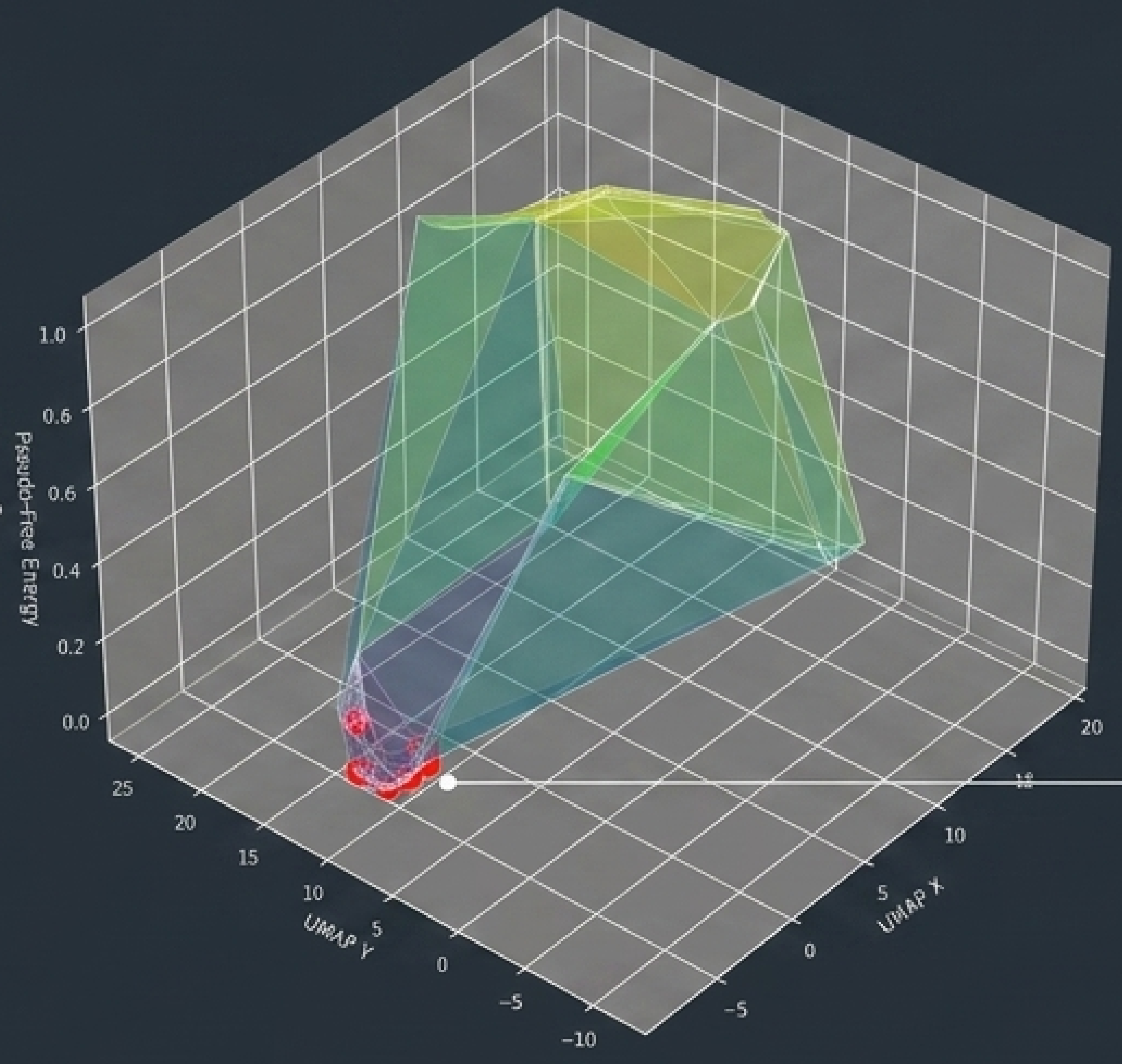
High-variance peripheral states. Demonstrates the broad representative capacity of the Behler-Parrinello transformation beyond standard local minima.

Core Takeaway Box

PCA confirms the transformation preserves deep chemical variance without structural collapse. The data contains rich, uncompressed informational content ready for algorithmic extraction.

State 2: Topological State Folding

Pseudo-Free Energy
The UMAP algorithm naturally forces the high-dimensional ACSF features into an elevation topology mimicking thermodynamic stability.



Discrete Chemical States
The invariant math naturally clusters specific structural conformers into defined, localized energy basins.

Core Takeaway Box

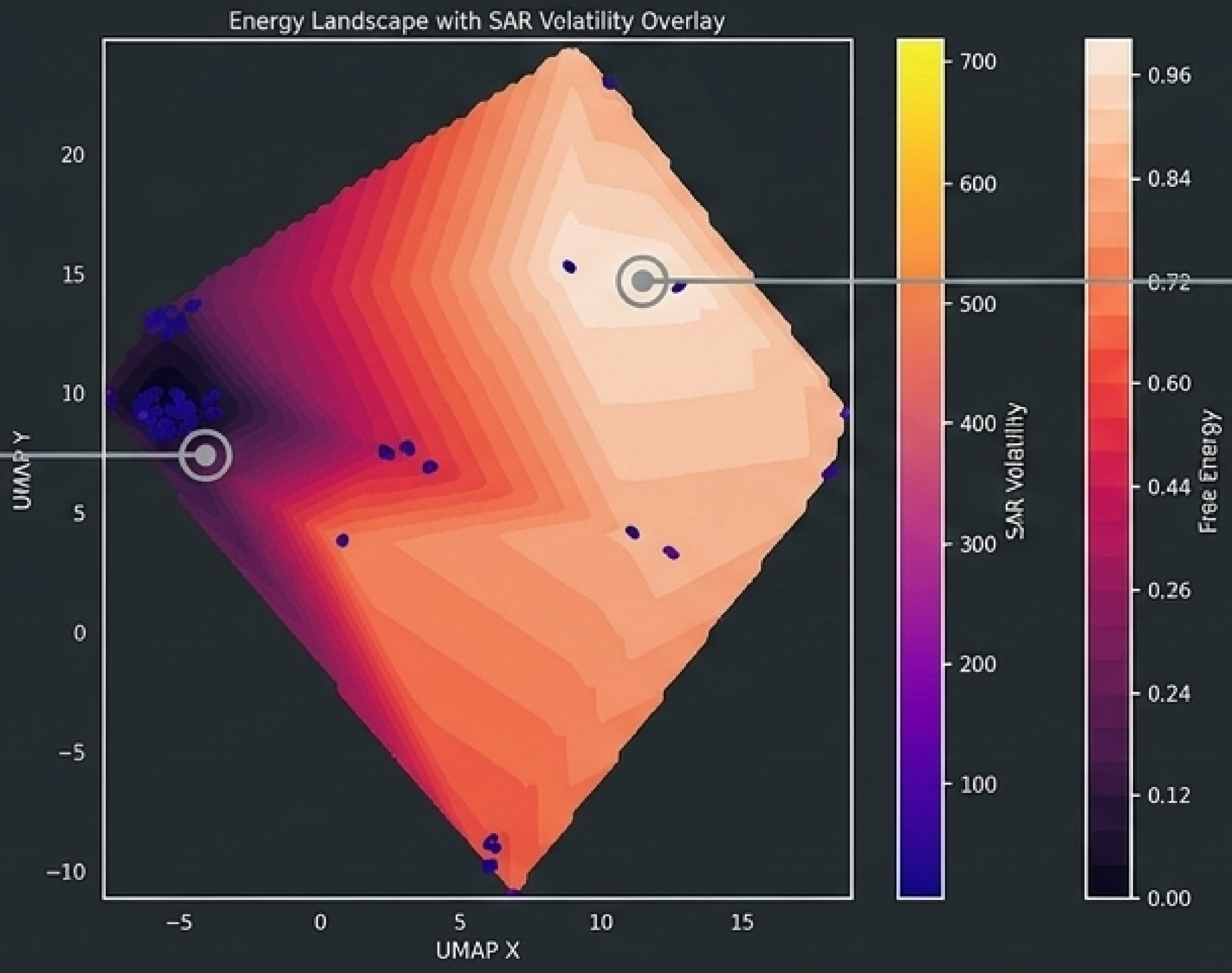
The dataset is not just mathematically invariant; it forces machine-learnable geometric clustering of distinct molecular conformations.

State 3: Physical Thermodynamics & Structure-Activity

1

Low-Energy Basins
[Free Energy ~0.00]

Highly stable, low-volatility conformations characterized by dense spatial clustering.



2

High-Volatility Gradients
[SAR Volatility > 600]

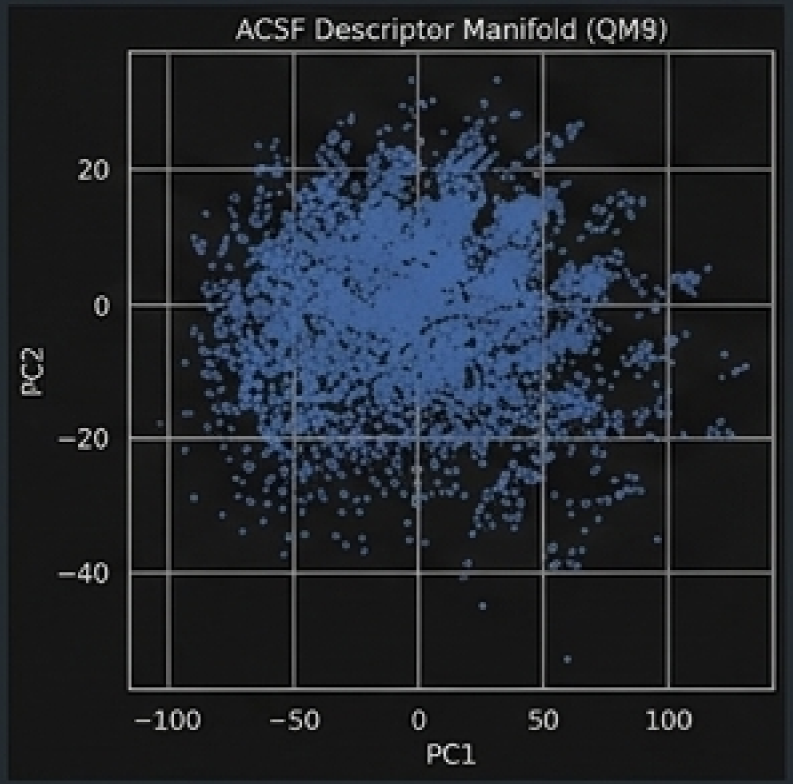
High-energy transition states mapping directly to highly active structural relationship zones.

Core Takeaway Box

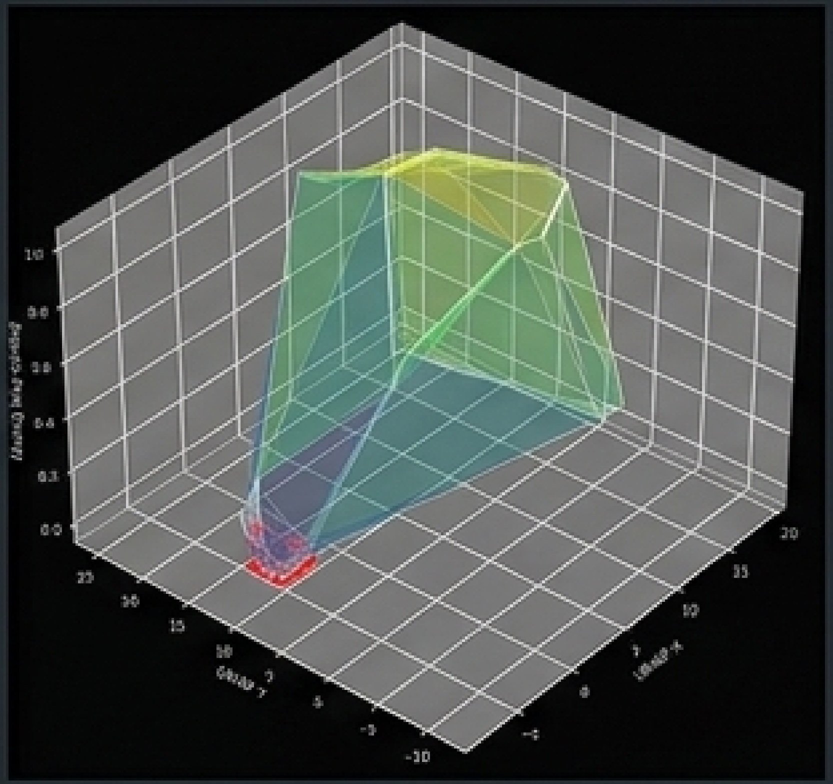
The ultimate validation: The engineered Iso-ACSF feature space mathematically aligns with actual quantum thermodynamic realities, perfectly setting up downstream surrogate modeling.

Synthesis: The Quantum-Mathematical Bridge

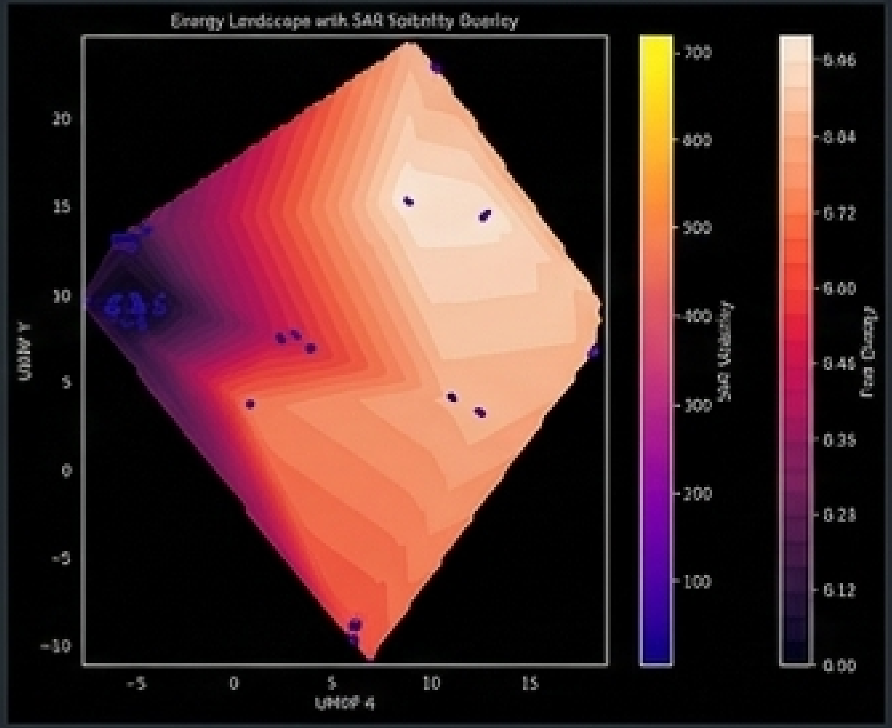
The Abstract



The Geometric

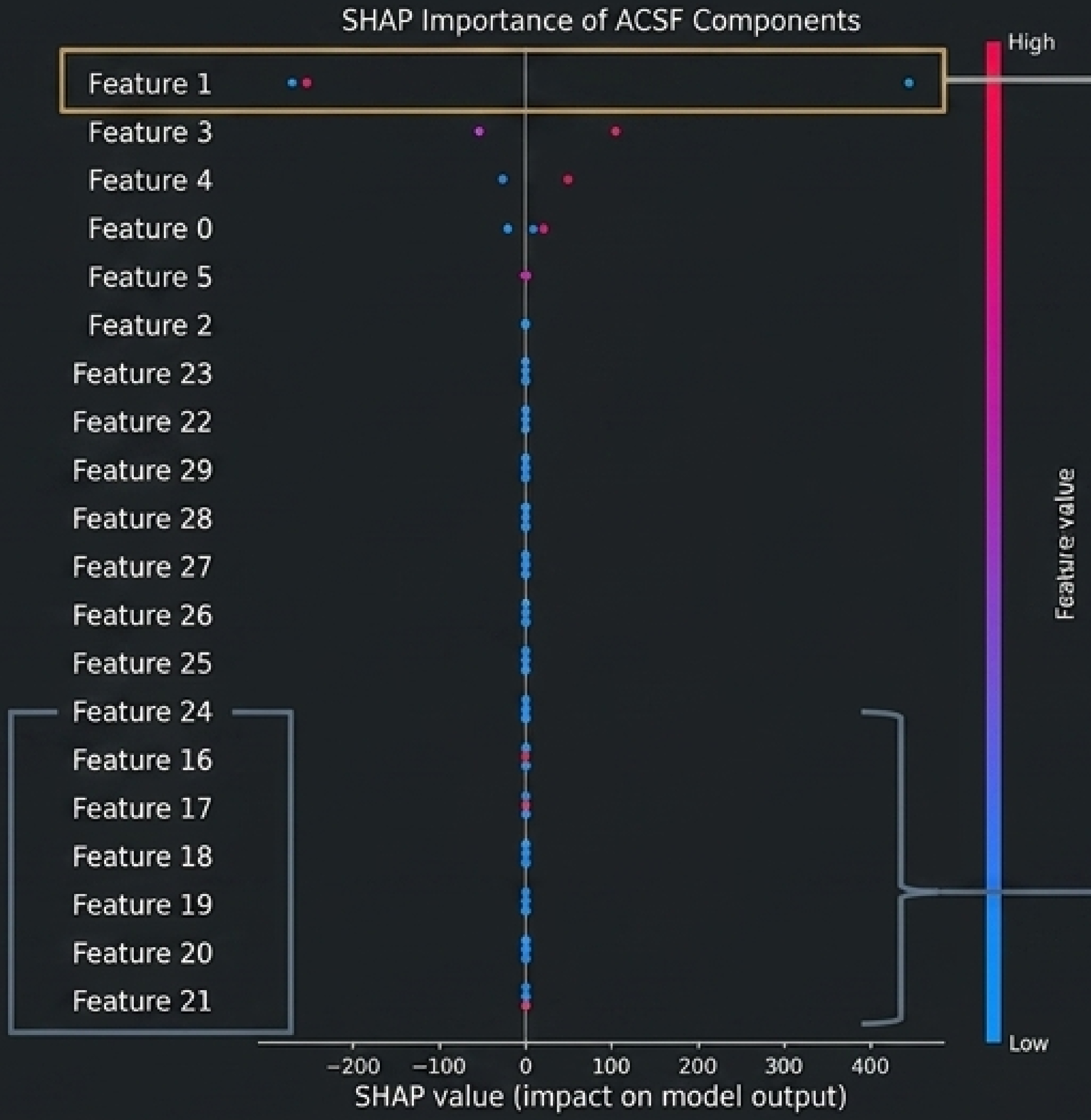


The Physical



Iso-ACSF is not merely a mathematical dimensionality trick. By tracking the data from raw Cartesian coordinates (PCA) through topological folding (UMAP) into localized energy minimums (SAR), we prove that this feature-engineered space perfectly mirrors quantum reality. Machine learning models trained on this manifold are learning foundational physics, not just statistical correlation.

Model Interpretability & Feature Mechanics

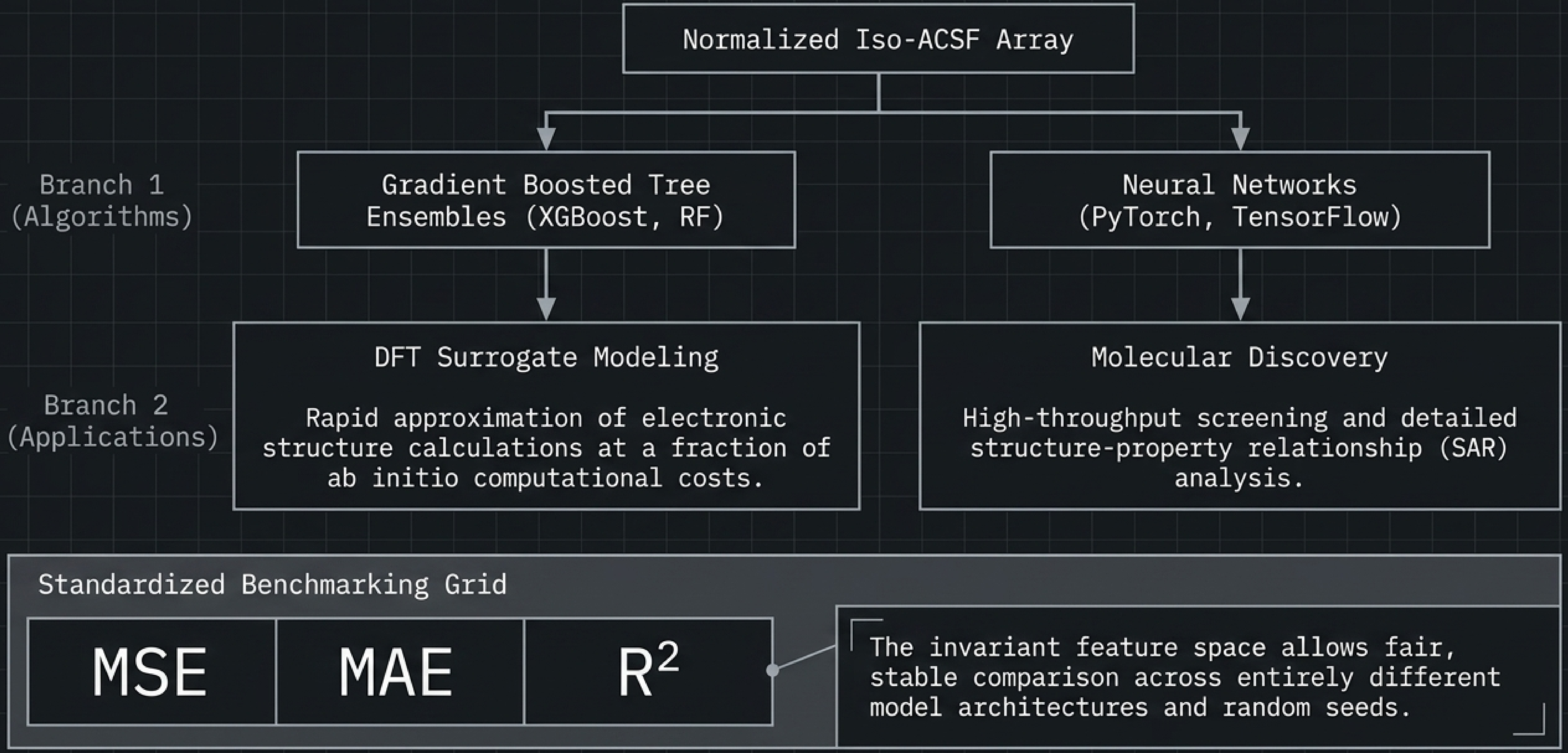


Primary Bifurcation Feature. The extreme bipolar dispersion (high vs. low impact) indicates this single ACSF dimension acts as a primary root node in tree-based regression, splitting major molecular structural classes.

Secondary Fine-Tuning Dimensions. These densely clustered, low-impact features provide the localized atomic environment granularity required for high-precision DFT approximations.

Core Takeaway Box
Iso-ACSF allows for fully transparent, physics-informed regression where specific symmetry functions can be mapped directly to model inference weight.

Target Implementations & Evaluation Architecture



Boundary Conditions & System Limitations

Spatial Opacity

No Explicit Coordinates

Because the data utilizes strictly invariant representations, it cannot be used for models requiring 3D spatial coordinate reconstruction or rigid-body docking.

Domain Specificity

QM9 Organic Constraint

The feature space and normalization parameters parameters are highly optimized exclusively for the small organic molecules natively represented in the QM9 benchmark.

Extrapolation Limits

Generalization Boundaries

Applying these specific normalized descriptors to drastically larger or radically distinct chemical systems requires from-scratch retraining or extensive feature adaptation.

System Access & Reproducibility Standards

Input: QM9 Geometries -> Function: Iso-ACSF -> Output: Highly Stable ML Surrogate Models

Compliance & Usage Protocol

1. Designed strictly for academic and industrial machine learning research.

2. Natively compliant with major structural ML frameworks (Scikit-learn, XGBoost, PyTorch).

3. Deterministic feature generation ensures absolute programmatic reproducibility.

Attribution Requirements

Users are required to cite the dataset via the Kaggle repository guidelines and officially acknowledge the original QM9 benchmark as the foundation for the molecular geometries.